第八周作业.

14.20. ~~$L=\frac{\Psi}{I}$~~

$2\pi r \cdot B = N \cdot I \mu_0 \quad B = \frac{NI\mu_0}{\pi r}$

$d\phi = B \cdot dS = \frac{NI}{2\pi r} \cdot dr \cdot h\mu_0$

$\Psi = \int N d\phi = \int_a^b \frac{NIh\mu_0}{2\pi r} \cdot dr \cdot N$

$= \frac{N^2 I h \mu_0}{2\pi} \cdot \ln\frac{b}{a}$

$L = \frac{\Psi}{I} = \frac{N^2 h \mu_0}{2\pi} \cdot \ln\frac{b}{a} = 1.39 \times 10^{-3}\ H/m.$

14.21.(1) $\bar{E} = -\frac{\Delta\Phi}{\Delta t}$

$B = \frac{\mu_0 n I}{2R} \qquad \phi = BS = \frac{\mu_0 n I}{2R} \cdot S$

$\Delta\phi = \frac{4\pi\times10^{-7}}{2\times0.2} \times 100 \times 20 \times 5\times10^{-4} = -3.14\times10^{-6}\ Wb$

$\bar{E} = -\frac{\Delta\phi}{\Delta t} = \frac{3.14\times10^{-6}}{0.01s} \cdot V = 3.14\times10^{-4} V.$

(2) $M = \frac{\Psi}{I} = \frac{\phi}{20A} = \frac{3.14\times10^{-6} Wb}{20}$

$= 1.57\times10^{-7}\ H.$

14.23. $B = \frac{\mu_0 I}{2\pi r} \quad d\phi = B \cdot dS = \frac{\mu_0 I}{2\pi r} (a+h-r) \cdot dr \frac{2}{\sqrt{3}}$

$\phi = \int d\phi = \int_a^{a+h} \frac{\mu_0 I}{\sqrt{3}\pi} \left(\frac{a+h}{r} - 1\right) dr = \frac{\mu_0 I}{\sqrt{3}\pi} \cdot \left(-h + (a+h) \cdot \ln\frac{a+h}{a}\right)$

则 $M = \frac{\phi}{I} = \frac{\mu_0}{\sqrt{3}\pi} \left[(a+h) \cdot \ln\frac{a+h}{a} - h\right]$

14.25. (1) $M=\frac{\phi}{I}$, $B=\frac{\mu_0 I}{2\pi r}$.

$$\phi=\int BdS=\int_{\frac{a}{2}}^{\frac{3a}{2}}\frac{\mu_0 I}{2\pi r}\cdot a\,dr=\frac{\mu_0 I a}{2\pi}\ln 3.$$

则 $M=\frac{\mu_0 a}{2\pi}\cdot\ln 3$

(2). $E=-M\frac{dI}{dt}=\frac{-\mu_0 a}{2\pi}\ln 3\cdot I_0\omega\cdot\cos\omega t$

$$=\frac{(-\ln 3)\mu_0 a\cdot I_0\omega}{2\pi}\cdot\cos\omega t.$$

~~解~~ 设在长直导线中通电流I.

14.26. $M=\frac{\Phi}{I}$ $B=\frac{\mu_0 I}{2\pi r}$. $dS=b\cdot dr$.

$$\Phi=\int BdS=\int_{R_1}^{R_2}\frac{\mu_0 I}{2\pi r}b\,dr=\frac{\mu_0 I b}{2\pi}\ln\frac{R_2}{R_1}.$$

则 $\psi=N\Phi$

$$M=\frac{\psi}{I}=\frac{\mu_0 N b}{2\pi}\ln\frac{R_2}{R_1}$$

~~(2)~~ $E=-M\cdot\frac{dI}{dt}=\frac{\mu_0 N b}{2\pi}\ln\frac{R_2}{R_1}\cdot I_0\cdot\omega\cdot\sin\omega t$.

$\omega t=\frac{\pi}{4}$ 时, $E=\frac{\sqrt{2}\mu_0 N b\cdot\omega I_0}{4\pi}\ln\frac{R_2}{R_1}=1.47\times10^{-2}V$.

15.2 (1) $I=\frac{d\Phi_D}{dt}=\frac{dE}{dt}\cdot\varepsilon_0\cdot\pi R^2=1\times10^{12}\times\pi\times(0.05)^2\times8.9\times10^{-12}$

$=0.07A$

(2). $2\pi r\cdot H=I$

则 $H=\frac{I}{2\pi r}$.

$B=\mu_0 H=4\pi\times10^{-7}\times\frac{0.07}{2\pi\times0.05}T=2.8\times10^{-7}T$.

15.3. (1). 细导线: $I_导=\frac{V}{R}=\frac{V_0\sin\omega t}{R}$.

(2) 位移电流: $I_位=\frac{d\Phi_D}{dt}=\frac{dE}{dt}\cdot A\cdot\varepsilon_0$.

$=\frac{d(\frac{V}{d})}{dt}\cdot\varepsilon_0 A=\frac{\varepsilon_0 A\omega V_0\cos\omega t}{d}$

(3). 外线电流: $I_总=I_导+I_位'$

$=\frac{V_0\sin\omega t}{R}+\frac{\varepsilon_0\cdot\omega V_0\cos\omega t\cdot\pi r^2}{d}$

(4). $2\pi r H=I_总$ 则 $H=\frac{I_总}{2\pi r}$.

$B=\mu_0 H=\frac{I_总\mu_0}{2\pi r}$

$=\cancel{\frac{}{2\pi rR}}\frac{V_0\sin\omega t}{2\pi r\cdot R}+\frac{\varepsilon_0\cdot\omega V_0\cos\omega t\cdot r}{2rd}$.

15.4 (1) $j = \frac{dD}{dt} = \frac{dE \cdot \varepsilon_0}{dt} = 8.9\times10^{-12} \times 720 \cdot \cos 10^5 \pi t \cdot 10^5 \pi$

$= 2\times10^{-3} \cos 10^5 \pi t.$

(2). $2\pi r \cdot H = I$

$I = j \cdot \pi r^2$，则 $H = \frac{I}{2\pi r} = \frac{j \cdot \pi r^2}{2\pi r} = \frac{jr}{2}$

$\cancel{H = \frac{\mu_0 j r}{2} = 3.2\times10^{-4} \cos 10^5 \pi t \cdot 4\pi\times10^{-7}\times10^{-2}}$

当 $t=0$ 时，$H = 1\times10^{-3}\times(\cos 10^5 \cdot \pi \cdot 0)\times10^{-2} = 1\times10^{-5}$ T.

当 $t = \frac{1}{2}\times10^{-5}$ 时，$H=0$

15.5 $I_{位} = \frac{d\Phi_D}{dt}$

$\Phi_D = D \cdot S = \int E\, dS \cdot \varepsilon_0 = \int_0^\theta \frac{kq}{\left(\frac{x}{\cos\alpha}\right)^2} (x\tan\alpha) \times 2\pi \frac{x\, d\alpha}{(\cos\alpha)^2} \cdot \varepsilon_0 \cos\alpha$

$= \int_0^\theta kq\, 2\pi \sin\alpha\, d\alpha\, \varepsilon_0$

$= (1-\cos\theta) \cdot kq\, 2\pi\varepsilon_0$

$= kq \cdot 2\pi\varepsilon_0 - \frac{x}{\sqrt{x^2+a^2}} \times \frac{q}{2}$

$I_{位} = \frac{d\Phi_D}{dt} = \frac{\sqrt{x^2+a^2}\cdot v - \frac{1}{2}\frac{2x\cdot v\cdot x}{\sqrt{x^2+a^2}}}{(x^2+a^2)} \left(\frac{q}{2}\right)$

$= \frac{(x^2+a^2)v - x^2 \cdot v}{(x^2+a^2)^{\frac{3}{2}}} \times \left(\frac{q}{2}\right)$

$= \frac{a^2 v q}{2(x^2+a^2)^{\frac{3}{2}}}$